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Vishwakarma Institute of Technology, Pune-37

(An autonomous Institute of Savitribai Phule Pune University)

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Title:
Study Assignment on Parallel and Distributed databases.

Description: In this assignment, I studied the concept of Parallel and Distributed Databases by selecting a real-world example and analyzing how it works in practice. I chose Google Spanner as the case study to understand how large-scale systems manage data across multiple servers located in different regions. The write-up explains how distributed databases improve performance, scalability, consistency, and fault tolerance using techniques like data sharding, replication, and parallel query execution. This study helped me understand why modern applications like banking systems, e-commerce platforms, and cloud services rely on distributed database architectures for high availability and global access.

* Study Assignment on Parallel & Distributed Databases:-

Google Spanner - A Real-World Distributed SQL Database

Parallel & Distributed Databases are designed to manage large-scale data across multiple machines instedd of relying on a single centralized system. Traditional databases struggle with performance & scalibility when the dataset become huge or users increase globally. Distributed systems solve this by splitting data across multiple servers (horizontal scaling) & performing parallel queries execution, which greatly improves speed, availability & fault tolerance.

Google spanner is a globally distributed relational database system built by Google. It powers many critical applications such as Gmail, Google Ads, Google Photos & YouTube. Unlike traditional database, Spanner combines consistency of SQL databases with scalability of NoSQL systems.

~~Arch~~ Architecture Overview:-

- Data Sharding (Partitioning):- Large tables are divided into smaller chunks stored across multiple servers.
- Replication :- Each chunk is copied 3+ times on different machines for fault tolerance.

- True Time API :- ~~[illegible]~~ Synchronizes global servers using GPS & atomic clocks to maintain accurate timestamps.
- Parallel Query Executor :- Queries are split & processed across multiple nodes simultaneously.

Key Advantages :-

- Scalibility :- more servers can be added instantly without downtime.
- High availability.
- Strong consistency
- Global data access.

How it performs Parallel + Distributed Operations :-

1) Instead of one powerful server, multiple smaller servers share the workload.
2) ~~where~~ When a query is executed, it is divided among many nodes → executed in parallel.
3) If one server fails, others automatically take over → ensuring zero downtime.
4) Write operations are coordinated using distributed consensus algorithm → ensuring data consistency across regions.

Conclusion:-

This study shows that Parallel & Distributed Databases are essential for modern large-scale applications. Google spanner is perfect example.

With the increasing demand for real-time data access across the world, such ~~data~~ database systems are becoming backbone of Cloud-based architecture. In the future, almost every large application will rely on distributed databases to ensure speed, reliability & scalibity.